

From Frame to Forward

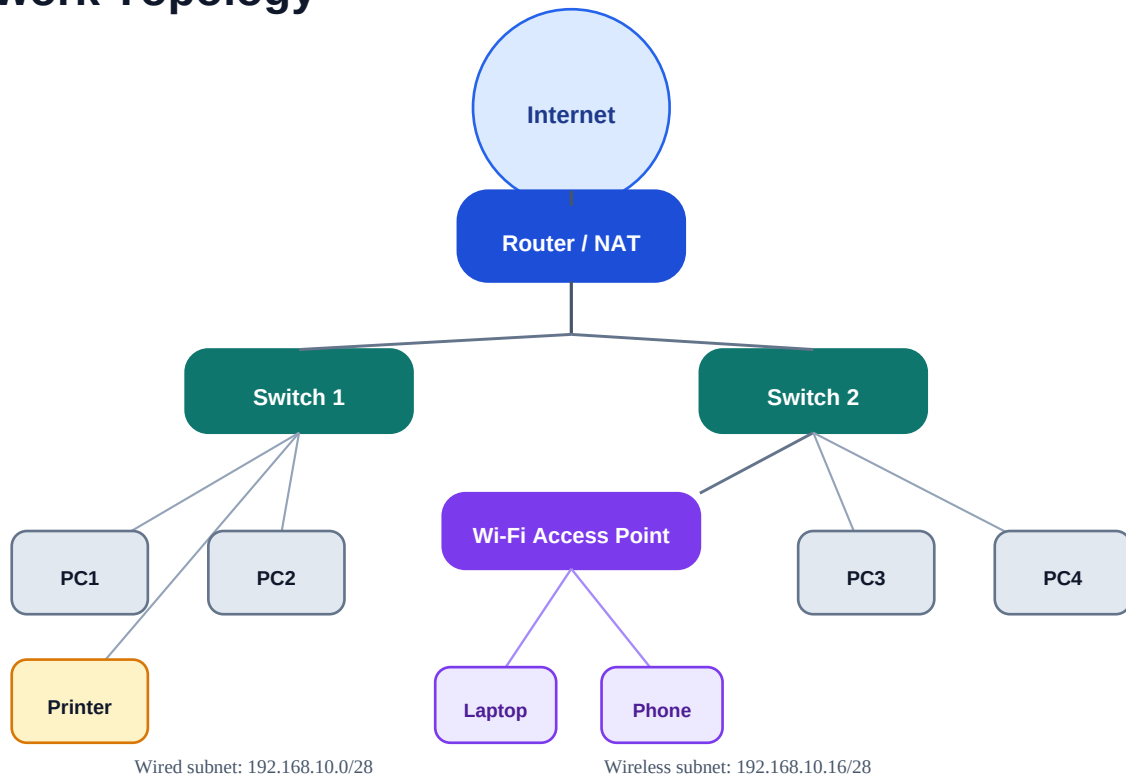
Building a Functional Local Network

Computer Systems and Their Fundamentals - Introduction to Computer Networks and Layers

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Scenario	Small company-floor LAN with wired, wireless, printer and internet access
Coverage	MAC addressing, Ethernet frames, collision handling, subnetting, DHCP, routing, forwarding and NAT

Design choice: the /24 address space is split into two /28 subnets to separate wired and wireless clients while keeping the design simple. The router provides inter-subnet routing, DHCP, NAT, and the internet default route.

Network Topology



The two switches serve the wired devices. The access point connects to the switching infrastructure and provides wireless access. The router is the Layer 3 boundary between local subnets and the internet.

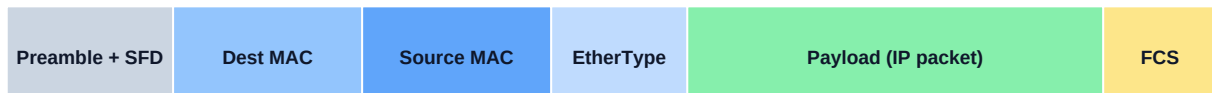
Part 1 - MAC Addressing and Frame Handling

Assigned MAC addresses

Device	Interface	MAC Address
Router	LAN/Wired	02:10:00:00:00:01
Router	LAN/Wireless	02:10:00:00:00:02
Switch 1	Management	02:10:00:00:10:01
Switch 2	Management	02:10:00:00:10:02
PC1	Ethernet	02:10:00:00:20:01
PC2	Ethernet	02:10:00:00:20:02
PC3	Ethernet	02:10:00:00:20:03
PC4	Ethernet	02:10:00:00:20:04
Printer	Ethernet	02:10:00:00:30:01
Wi-Fi AP	Wireless/LAN	02:10:00:00:40:01
Laptop	Wi-Fi	02:10:00:00:50:01
Phone	Wi-Fi	02:10:00:00:50:02

Ethernet frame from PC1 to the printer

Ethernet II Frame - PC1 to Printer



Before PC1 can send the Ethernet frame, it needs the printer's MAC address. Because PC1 and the printer are in the same wired subnet, PC1 uses ARP to map the printer's IP address to its MAC address. The resulting frame contains the printer MAC as the destination MAC and PC1's MAC as the source MAC. The payload carries the Layer 3 IP packet. The FCS is used by the receiver to detect transmission errors.

Field	Example / Purpose
Destination MAC	02:10:00:00:30:01 - Printer
Source MAC	02:10:00:00:20:01 - PC1
EtherType	0x0800 for IPv4
Payload	IP packet containing the print/application data
FCS	CRC-based error-detection value

Collision handling

Wired Ethernet: With modern switched full-duplex Ethernet, each switch port is its own collision domain and the endpoint can transmit and receive at the same time. Therefore collisions are effectively eliminated on normal full-duplex links. CSMA/CD is mainly relevant to older shared-media or half-duplex Ethernet.

Wi-Fi: Wireless devices share the radio medium and cannot reliably detect collisions while transmitting. Wi-Fi therefore uses CSMA/CA: a device senses the channel, waits if it is busy, uses randomized backoff before transmission, and relies on acknowledgements. Optional RTS/CTS can reduce hidden-node collisions.

Part 2 - IP Addressing and Subnetting

Private range selected: **192.168.10.0/24**. It is divided into two /28 subnets. Each /28 has 16 addresses: 1 network address, 14 usable host addresses, and 1 broadcast address.

Purpose	Subnet	Usable Range	Gateway	Broadcast
Wired LAN	192.168.10.0/28	192.168.10.1 - 192.168.10.14	192.168.10.1	192.168.10.15
Wireless LAN	192.168.10.16/28	192.168.10.17 - 192.168.10.30	192.168.10.17	192.168.10.31

IP address assignment

Device	IP Address	Mask	Default Gateway
Router - Wired	192.168.10.1	255.255.255.240	N/A
Switch 1 mgmt	192.168.10.2	255.255.255.240	192.168.10.1
Switch 2 mgmt	192.168.10.3	255.255.255.240	192.168.10.1
PC1	192.168.10.4	255.255.255.240	192.168.10.1
PC2	192.168.10.5	255.255.255.240	192.168.10.1
PC3	192.168.10.6	255.255.255.240	192.168.10.1
PC4	192.168.10.7	255.255.255.240	192.168.10.1
Printer	192.168.10.8	255.255.255.240	192.168.10.1
Router - Wireless	192.168.10.17	255.255.255.240	N/A
Wi-Fi AP mgmt	192.168.10.18	255.255.255.240	192.168.10.17
Laptop	192.168.10.19	255.255.255.240	192.168.10.17
Phone	192.168.10.20	255.255.255.240	192.168.10.17

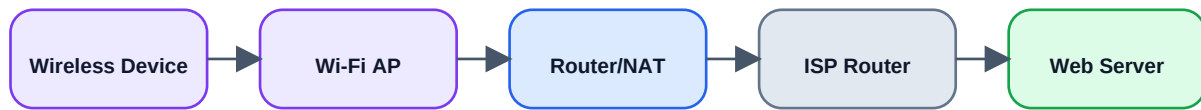
Role of DHCP

DHCP automatically supplies hosts with IP configuration. In this design, the router can operate two DHCP pools: one for the wired /28 and one for the wireless /28. A client uses the DHCP DORA sequence - Discover, Offer, Request, Acknowledge - to obtain an address, subnet mask, default gateway, lease duration, and DNS server information.

Infrastructure devices such as the router interfaces, switches, access point, and printer are better assigned fixed addresses or DHCP reservations so that their management or service addresses do not change.

Part 3 - Routing and Packet Forwarding

Outbound packet path to an external website



Private IP -> NAT public IP -> Internet -> Destination

Example: wireless laptop accesses an external website

Step	What happens
1	The laptop resolves the website name using DNS and obtains the server's public IP address.
2	The laptop compares the destination IP with its own subnet 192.168.10.16/28. Because the website is outside the local subnet, it sends
3	The laptop uses ARP to learn the MAC address of the gateway if it is not already cached.
4	The laptop creates an IP packet with source IP 192.168.10.19 and destination IP = website public IP, then encapsulates it in a Wi-Fi frame
5	The access point bridges the wireless traffic into the LAN. The router receives the Layer 3 packet.
6	The router reads the destination IP in the IP header and checks its routing table for the best matching route. The default route 0.0.0.0/0
7	Before forwarding to the ISP, the router applies NAT/PAT: source 192.168.10.19:source-port is translated to the router's public IP plus a
8	Each router on the internet examines the destination IP, decrements TTL, updates the IPv4 header checksum, and forwards the packet
9	The website server replies to the router's public IP. The company router uses the NAT translation table to map the response back to 192

How the IP header is used

Routers primarily use the **destination IP address** to decide where a packet should go. The source IP identifies the sender, while fields such as TTL prevent indefinite routing loops. The router does not normally preserve the original Layer 2 frame across hops: at every routed hop, the old frame is removed and a new Layer 2 frame is built for the next link, while the Layer 3 destination generally remains the final server address.

How the router decides the next hop

The router compares the destination IP address against entries in its routing table and chooses the **longest-prefix match**. Connected routes handle the local wired and wireless subnets. External traffic uses the default route toward the ISP. The selected route determines the outgoing interface and, where required, the next-hop router.

Role of NAT

The LAN uses private addresses that are not routed directly on the public internet. NAT translates the internal source address to the router's public address. With PAT, many internal hosts can share that single public IP by using different transport-layer port mappings. Return traffic is matched against the NAT table and translated back to the correct internal host.

End-to-End Layer View

Layer	Inside local LAN	At router boundary	Across Internet
Application	Browser / print app generates data	Payload unchanged	Server application receives data
Transport	TCP/UDP ports identify sessions	PAT may translate source port	Ports identify end-to-end flow
Network	Private source IP + destination IP	Routing decision + NAT	Routers forward based on destination IP
Data Link	Ethernet or 802.11 MAC frame	Old frame removed; new frame created	Different Layer 2 framing per hop
Physical	Copper/radio bits	WAN medium	Fiber/copper/wireless carrier links

Final Summary

Topic	Design Result
MAC	Unique locally administered MAC addresses assigned to all interfaces/devices.
Frames	PC-to-printer traffic uses destination/source MAC, EtherType, payload and FCS.
Collisions	Switched full-duplex Ethernet avoids collisions; Wi-Fi uses CSMA/CA.
Subnets	Wired = 192.168.10.0/28; Wireless = 192.168.10.16/28.
DHCP	Router can dynamically lease IP, mask, gateway and DNS settings.
Routing	Router uses destination IP and longest-prefix matching; default route points to ISP.
NAT	Private addresses are translated to a public IP, typically using PAT.

This design provides a small but functional company-floor LAN and demonstrates the transition from local Layer 2 frame delivery to Layer 3 routing and internet forwarding.